

# Kernel machines - PART II

September, 2008  
MLSS'08

Laboratoire d'Informatique, de Traitement de l'Information et des Systèmes

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# In the beginning was the kernel...

## Definition (Kernel)

a function of two variable  $k$  from  $\mathcal{X} \times \mathcal{X}$  to  $\mathbb{R}$

## Definition (Positive kernel)

A kernel  $k(s, t)$  on  $\mathcal{X}$  is said to be positive

- ▶ if it is symmetric:  $k(s, t) = k(t, s)$
- ▶ and if for any finite positive integer  $n$ :

$$\forall \{\alpha_i\}_{i=1,n} \in \mathbb{R}, \forall \{\mathbf{x}_i\}_{i=1,n} \in \mathcal{X}, \quad \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0$$

it is strictly positive if for  $\alpha_i \neq 0$

$$\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) > 0$$



# Examples of positive kernels

**the linear kernel:**  $\mathbf{s}, \mathbf{t} \in \mathbb{R}^d$ ,  $k(\mathbf{s}, \mathbf{t}) = \mathbf{s}^\top \mathbf{t}$

symetric:  $\mathbf{s}^\top \mathbf{t} = \mathbf{t}^\top \mathbf{s}$

$$\begin{aligned} \text{positive: } \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \mathbf{x}_i^\top \mathbf{x}_j \\ &= \left( \sum_{i=1}^n \alpha_i \mathbf{x}_i \right)^\top \left( \sum_{j=1}^n \alpha_j \mathbf{x}_j \right) = \left\| \sum_{i=1}^n \alpha_i \mathbf{x}_i \right\|^2 \end{aligned}$$

**the product kernel:**  $k(\mathbf{s}, \mathbf{t}) = g(\mathbf{s})g(\mathbf{t})$  for some  $g : \mathbb{R}^d \rightarrow \mathbb{R}$ ,

symetric by construction

$$\begin{aligned} \text{positive: } \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j g(\mathbf{x}_i) g(\mathbf{x}_j) \\ &= \left( \sum_{i=1}^n \alpha_i g(\mathbf{x}_i) \right) \left( \sum_{j=1}^n \alpha_j g(\mathbf{x}_j) \right) = \left( \sum_{i=1}^n \alpha_i g(\mathbf{x}_i) \right)^2 \end{aligned}$$

$k$  is positive  $\Leftrightarrow$  (its square root exists)  $\Leftrightarrow k(\mathbf{s}, \mathbf{t}) = \langle \phi_{\mathbf{s}}, \phi_{\mathbf{t}} \rangle$





## positive definite Kernel (PDK) algebra (closure)

if  $k_1(\mathbf{s}, \mathbf{t})$  and  $k_2(\mathbf{s}, \mathbf{t})$  are two positive kernels

- ▶ DPK are a convex cone:  $\forall a_1 \in \mathbb{R}^+ \quad a_1 k_1(\mathbf{s}, \mathbf{t}) + k_2(\mathbf{s}, \mathbf{t})$
- ▶ for any measurable function  $\psi$  from  $\mathcal{X}$  to  $\mathbb{R}$   $k(\mathbf{s}, \mathbf{t}) = \psi(\mathbf{s})\psi(\mathbf{t})$
- ▶ product kernel  $k_1(\mathbf{s}, \mathbf{t})k_2(\mathbf{s}, \mathbf{t})$

### proofs

- ▶ by linearity:

$$\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j (a_1 k_1(i, j) k_2(i, j)) = a_1 \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k_1(i, j) + \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k_2(i, j)$$

- ▶ by linearity:  $\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j (\psi(\mathbf{x}_i) \psi(\mathbf{x}_j)) = \left( \sum_{i=1}^n \alpha_i \psi(\mathbf{x}_i) \right) \left( \sum_{j=1}^n \alpha_j \psi(\mathbf{x}_j) \right)$

- ▶ assuming  $\exists \psi_\ell$  s.t.  $k_1(\mathbf{s}, \mathbf{t}) = \sum_\ell \psi_\ell(\mathbf{s}) \psi_\ell(\mathbf{t})$

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k_1(\mathbf{x}_i, \mathbf{x}_j) k_2(\mathbf{x}_i, \mathbf{x}_j) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \left( \sum_\ell \psi_\ell(\mathbf{x}_i) \psi_\ell(\mathbf{x}_j) k_2(\mathbf{x}_i, \mathbf{x}_j) \right) \\ &= \sum_\ell \sum_{i=1}^n \sum_{j=1}^n (\alpha_i \psi_\ell(\mathbf{x}_i)) (\alpha_j \psi_\ell(\mathbf{x}_j)) k_2(\mathbf{x}_i, \mathbf{x}_j) \end{aligned}$$



# Kernel engineering: building PDK

- ▶ for any polynomial with positive coef.  $\phi$  from  $\mathbb{R}$  to  $\mathbb{R}$   
 $\phi(k(s, t))$
- ▶ if  $\Psi$  is a function from  $\mathbb{R}^d$  to  $\mathbb{R}^d$   
 $k(\Psi(s), \Psi(t))$
- ▶ if  $\varphi$  from  $\mathbb{R}^d$  to  $\mathbb{R}^+$ , is minimum in 0  
 $k(s, t) = \varphi(s + t) - \varphi(s - t)$
- ▶ convolution of two positive kernels is a positive kernel  
 $K_1 \star K_2$

## the Gaussian kernel is a PDK

$$\begin{aligned} \exp(-\|s - t\|^2) &= \exp(-\|s\|^2 - \|t\|^2 - 2s^\top t) \\ &= \exp(-\|s\|^2) \exp(-\|t\|^2) \exp(2s^\top t) \end{aligned}$$

- ▶  $s^\top t$  is a PDK and function  $\exp$  as the limit of positive series expansion, so  $\exp(2s^\top t)$  is a PDK
- ▶  $\exp(-\|s\|^2) \exp(-\|t\|^2)$  is a PDK as a product kernel
- ▶ the product of two PDK is a PDK



# an attempt at classifying PD kernels

- ▶ stationary kernels, (also called translation invariant):

$$k(s, t) = k_s(s - t)$$

- ▶ radial (isotropic)                      gaussian:  $\exp\left(-\frac{r^2}{b}\right)$ ,  $r = \|s - t\|$
- ▶ with compact support

$$\text{c.s. Matérn : } \max\left(0, 1 - \left(\frac{r}{b}\right)^\kappa\right) \frac{r^\kappa}{b} B_\kappa\left(\frac{r}{b}\right), \quad \kappa \geq (d + 1)/2$$

- ▶ locally stationary kernels:                       $k(s, t) = k_1(s + t)k_s(s - t)$   
 $K_1$  is a non negative function and  $K_2$  a radial kernel.

- ▶ non stationary (projective kernels):

$$k(s, t) = k_p(s^\top t)$$

- ▶ separable kernels  $k(s, t) = k_1(s)k_2(t)$  with  $k_1$  and  $k_2(t)$  PDK  
 in this case  $K = k_1 k_2^\top$  where  $k_1 = (k_1(\mathbf{x}_1), \dots, k_1(\mathbf{x}_n))$



# Kernel spectral representation

- ▶ stationary kernels, (Bochner's theorem):

$$k_s(\mathbf{s} - \mathbf{t}) = \int_{\mathbf{R}^d} \cos(\omega^\top (\mathbf{s} - \mathbf{t})) F(d\omega)$$

with  $F$  a **positive** finite measure

- ▶ radial kernels

$$k_r(\|\mathbf{s} - \mathbf{t}\|) = \int_0^\infty \Psi_d(\omega \|\mathbf{s} - \mathbf{t}\|) F(d\omega)$$

where  $F$  is non decreasing and bounded, and  $\Psi_d$  a specific function

- ▶ non stationary (projective kernels):

$$k_p(s, t) = \int_{\mathbf{R}^d} \int_{\mathbf{R}^d} \cos(\omega_1^\top \mathbf{s} - \omega_2^\top \mathbf{t}) F(d\omega_1, d\omega_2)$$

with  $F$  a positive bounded symmetric measure

**Fourier transform may help to design PD kernels**









# kernels for objects and structures

kernels on histograms and probability distributions

$$k(p(x), q(x)) = \int k_i(p(x), q(x)) \mathbb{P}(x) dx$$

kernel on strings

- ▶ spectral string kernel

$$k(\mathbf{s}, \mathbf{t}) = \sum_u \phi_u(\mathbf{s}) \phi_u(\mathbf{t})$$

- ▶ using sub sequences

- ▶ similarities by alignments

$$k(\mathbf{s}, \mathbf{t}) = \sum_{\pi} \exp(\beta(\mathbf{s}, \mathbf{t}, \pi))$$

kernels on graphs

- ▶ the pseudo inverse of the (regularized) graph Laplacian

$$L = D - A \quad A \text{ is the adjacency matrix } D \text{ the degree matrix}$$

- ▶ diffusion kernels

$$\frac{1}{Z(b)} \exp^{bL}$$

- ▶ subgraph kernel convolution (using random walks)

and kernels on heterogeneous data (image), HMM, automata...



# Gram matrix

## Definition (Gram matrix)

let  $k(\mathbf{s}, \mathbf{t})$  be a positive kernel on  $\mathcal{X}$  and  $(\mathbf{x}_i)_{i=1,n}$  a sequence on  $\mathcal{X}$ .  
the Gram matrix is the square  $K$  of dimension  $n$  and of general term  
 $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ .

practical trick to check kernel positivity:

$K$  is positive  $\Leftrightarrow \lambda_i > 0$  its eigenvalues are positives: if

$$K \mathbf{u}_i = \lambda_i \mathbf{u}_i; i = 1, n$$

$$\mathbf{u}_i^\top K \mathbf{u}_i = \lambda_i \mathbf{u}_i^\top \mathbf{u}_i = \lambda_i$$

matrix  $K$  is the one to be used





# different point of view about kernels

kernel and scalar product

$$k(\mathbf{s}, \mathbf{t}) = \langle \phi(\mathbf{s}), \phi(\mathbf{t}) \rangle_{\mathcal{H}}$$

kernel and distance

$$d(\mathbf{s}, \mathbf{t})^2 = k(\mathbf{s}, \mathbf{s}) + k(\mathbf{t}, \mathbf{t}) - 2k(\mathbf{s}, \mathbf{t})$$

kernel and covariance: a positive matrix is a covariance matrix

$$\mathbb{P}(\mathbf{f}) = \frac{1}{Z} \exp\left(-\frac{1}{2}(\mathbf{f} - \mathbf{f}_0)^\top K^{-1}(\mathbf{f} - \mathbf{f}_0)\right)$$

$$\text{if } \mathbf{f}_0 = 0 \text{ and } \mathbf{f} = K\boldsymbol{\alpha}, \mathbb{P}(\boldsymbol{\alpha}) = \frac{1}{Z} \exp - \frac{1}{2} \boldsymbol{\alpha}^\top K \boldsymbol{\alpha}$$

Kernel and regularity (green's function)

$$k(\mathbf{s}, \mathbf{t}) = P^* P \delta_{\mathbf{s}-\mathbf{t}} \quad \text{for some operator } P \quad (\text{e.g. some differential})$$

# Let's summarize

- ▶ positive kernels
- ▶ there is a lot of them
- ▶ can be rather complex
- ▶ 2 classes: radial / projective
- ▶ the bandwidth matters (more than the kernel itself)
- ▶ the Gram matrix summarizes the pairwise comparisons





# From kernel to functions

$$\mathcal{H}_0 = \left\{ f \mid m_f < \infty; f_j \in \mathbb{R}; t_j \in \mathcal{X}, f(\mathbf{x}) = \sum_{j=1}^{m_f} f_j k(\mathbf{x}, t_j) \right\}$$

let define the bilinear form ( $g(\mathbf{x}) = \sum_{i=1}^{m_g} g_i k(\mathbf{x}, s_i)$ ) :

$$\forall f, g \in \mathcal{H}_0, \langle f, g \rangle_{\mathcal{H}_0} = \sum_{j=1}^{m_f} \sum_{i=1}^{m_g} f_j g_i k(t_j, s_i)$$

**Evaluation functional:**  $\forall \mathbf{x} \in \mathcal{X}$

$$f(\mathbf{x}) = \langle f(\cdot), k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}_0}$$

**from  $k$  to  $\mathcal{H}$**

with any positive kernel, a hypothesis set can be constructed  $\mathcal{H}$  with its metric

# RKHS

## Definition (reproducing kernel Hilbert space (RKHS))

a Hilbert space  $\mathcal{H}$  embedded with the inner product  $\langle \cdot, \cdot \rangle_{\mathcal{H}}$  is said to be with reproducing kernel if it exists a positive kernel  $k$  such that

$$\forall s \in \mathcal{X}, k(\cdot, s) \in \mathcal{H} \text{ et } \forall f \in \mathcal{H}, \quad f(s) = \langle f(\cdot), k(s, \cdot) \rangle_{\mathcal{H}}$$

## positive kernel $\Leftrightarrow$ RKHS

- ▶ any function is pointwise defined
- ▶ defines the inner product
- ▶ it defines the **regularity** (smoothness) of the hypothesis set



# functional differentiation in RKHS

Let  $J$  be a functional

$$J: \mathcal{H} \rightarrow \mathbb{R} \quad \text{examples:} \quad J_1(f) = \|f\|^2, J_2(f) = f(\mathbf{x}),$$

$$f \mapsto J(f)$$

$J$  directional derivative in direction  $g$  at point  $f$

$$dJ(f, g) = \lim_{\varepsilon \rightarrow 0} \frac{J(f + \varepsilon g) - J(f)}{\varepsilon}$$

Gradient  $\nabla_J(f)$

$$\begin{array}{ccc} \nabla_J: \mathcal{H} & \rightarrow & \mathcal{H} \\ f & \mapsto & \nabla_J(f) \end{array} \quad \text{si} \quad dJ(f, g) = \langle \nabla_J(f), g \rangle_{\mathcal{H}}$$

exercice: find out  $\nabla_{J_1}(f)$  et  $\nabla_{J_2}(f)$



# Hint

$$dJ(f, g) = \left. \frac{dJ(f + \varepsilon g)}{d\varepsilon} \right|_{\varepsilon=0}$$



# Solution

$$\begin{aligned}
 dJ_1(f, g) &= \lim_{\varepsilon \rightarrow 0} \frac{\|f + \varepsilon g\|^2 - \|f\|^2}{\varepsilon} \\
 &= \lim_{\varepsilon \rightarrow 0} \frac{\|f\|^2 + \varepsilon^2 \|g\|^2 + 2\varepsilon \langle f, g \rangle_{\mathcal{H}} - \|f\|^2}{\varepsilon} \\
 &= \lim_{\varepsilon \rightarrow 0} \varepsilon \|g\|^2 + 2\langle f, g \rangle_{\mathcal{H}} \quad \Leftrightarrow \quad \nabla_{J_1}(f) = 2f \\
 &= \langle 2f, g \rangle_{\mathcal{H}}
 \end{aligned}$$

$$\begin{aligned}
 dJ_2(f, g) &= \lim_{\varepsilon \rightarrow 0} \frac{f(\mathbf{x}) + \varepsilon g(\mathbf{x}) - f(\mathbf{x})}{\varepsilon} \\
 &= g(\mathbf{x}) \quad \Leftrightarrow \quad \nabla_{J_2}(f) = k(\mathbf{x}, \cdot) \\
 &= \langle k(\mathbf{x}, \cdot), g \rangle_{\mathcal{H}}
 \end{aligned}$$

$$\text{Minimize}_{f \in \mathcal{H}} J(f) \quad \Leftrightarrow \quad \forall g \in \mathcal{H}, dJ(f, g) = 0 \quad \Leftrightarrow \quad \nabla_J(f) = 0$$



# Solution

$$\begin{aligned}
 dJ_1(f, g) &= \lim_{\varepsilon \rightarrow 0} \frac{\|f + \varepsilon g\|^2 - \|f\|^2}{\varepsilon} \\
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 \end{aligned}$$

$$\text{Minimize}_{f \in \mathcal{H}} J(f) \quad \Leftrightarrow \quad \forall g \in \mathcal{H}, dJ(f, g) = 0 \quad \Leftrightarrow \quad \nabla_J(f) = 0$$

## Remark about the regularity matter: from $\mathcal{H}$ to $k$

How to build  $\mathcal{H}$  together with  $k$ ?

let  $\Gamma_t$  be a family in  $L^2$ ,  $t \in \mathcal{X}$

define the following mapping

$$S_t: L^2 \rightarrow \mathbb{R}$$

$$g \mapsto S_t(g) = \int g(x) \Gamma_t(x) dx \quad f(t) = S_t(g) = \langle g, \Gamma_t \rangle_{L^2}$$

the following RKHS can be constructed

$$\mathcal{H} = \text{Im}(S) \quad \langle f_1, f_2 \rangle_{\mathcal{H}} = \langle g_1, g_2 \rangle_{L^2} \quad k(s, t) = \langle \Gamma_s, \Gamma_t \rangle_{L^2}$$

the reproducing property is verified:

$$\langle f(\cdot), k(\cdot, t) \rangle_{\mathcal{H}} = \langle g, \Gamma_t \rangle_{L^2} = f(t)$$



# $\Gamma_x$ examples and associated kernels

|                | $\Gamma_x(u)$                                     | $K(x, y)$                                         |
|----------------|---------------------------------------------------|---------------------------------------------------|
| Cameron Martin | $\mathbb{I}_{\{x \leq u\}}$                       | $\min(x, y)$                                      |
| Polynomial     | $e_0(u) + \sum_{i=1}^d x_i e_i(u)$                | $\mathbf{x}^\top \mathbf{y} + 1$                  |
| Gaussian       | $\frac{1}{\sqrt{2\pi}} \exp^{-\frac{(x-u)^2}{2}}$ | $\frac{1}{\sqrt{2\pi}} \exp^{-\frac{(x-y)^2}{2}}$ |

$\{e_i\}_{i=1,d}$  are a finite sub sample of an orthonormal basis in  $L^2$

the Cameron Martin operator:  $\forall f \in \mathcal{H}, \exists g \in L^2$  such that

$$f(t) = (Sg)(t) = \int \Gamma_t(u) g(u) du = \int \mathbb{I}_{\{t \leq u\}} g(u) du = G(t)$$

$$\|f\|_{\mathcal{H}} = \|Sg\|_{\mathcal{H}} = \|g\|_{L^2}^2 = \|f'\|_{L^2}^2 \quad \text{generalized differential}$$

integrate  $S = P^{-1}$  differentialize



# $\Gamma_x$ examples and associated kernels

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| Gaussian       | $\frac{1}{Z} \exp^{-\frac{(x-u)^2}{2}}$ | $\frac{1}{Z'} \exp^{-\frac{(x-y)^2}{4}}$ |

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$$\|f\|_{\mathcal{H}} = \|Sg\|_{\mathcal{H}} = \|g\|_{L^2}^2 = \|f'\|_{L^2}^2 \quad \text{generalized differential}$$

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## other kernels (what really matters)

- ▶ finite kernels

$$k(\mathbf{s}, \mathbf{t}) = (\phi_1(\mathbf{s}), \dots, \phi_p(\mathbf{s}))^\top (\phi_1(\mathbf{t}), \dots, \phi_p(\mathbf{t}))$$

- ▶ Mercer kernels

positive on a compact set  $\Leftrightarrow k(\mathbf{s}, \mathbf{t}) = \sum_{j=1}^p \lambda_j \phi_j(\mathbf{s}) \phi_j(\mathbf{t})$

- ▶ **positive kernels**

- ▶ positive semi-definite

- ▶ conditionnaly positive (for some functions  $p_j$ )

$$\forall \{\mathbf{x}_i\}_{i=1,n}, \forall \alpha_i, \sum_i^n \alpha_i p_j(\mathbf{x}_i) = 0; \quad j = 1, p, \quad \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0$$

- ▶ symetric non positive

$$k(\mathbf{s}, \mathbf{t}) = \tanh(\mathbf{s}^\top \mathbf{t} + \alpha_0)$$

- ▶ non symetric – non positive

the key property:  $\nabla_{J_t}(f) = k(t, \cdot)$  holds



# Let's summarize

- ▶ positive kernels  $\Leftrightarrow$  RKHS  $= \mathcal{H} \Leftrightarrow$  regularity  $\|f\|_{\mathcal{H}}^2$
- ▶ the key property:  $\nabla_{J_t}(f) = k(t, \cdot)$  holds not only for positive kernels  
 $f(\mathbf{x}_i)$  exists (pointwise defined functions)
- ▶ universal consistency in RKHS
- ▶ the Gram matrix summarize the pairwise comparisons

# Roadmap

## 1 Kernels and the learning problem

- Three learning problems
- Learning from data: the problem
- Kernelizing the linear regression
- Kernel machines: a definition

## 2 Tools: the functional framework

- In the beginning was the kernel
- Kernel and hypothesis set
- Optimization, loss function and the regularization cost

## 3 Kernel machines

- Non sparse kernel machines
- Sparse kernel machines
- practical SVM

## 4 Conclusion

# Convex optimization in one slide

- The problem:  $\mathcal{H}$  is a Hilbert space ( $\mathbb{R}^n$  or a RKHS)

$$OP \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} J(f) & \text{objective} \\ \text{such that } P_i(f) \leq 0 \quad i = 1, p & \text{constraint} \\ \text{and } R_j(f) = 0 \quad j = 1, q \end{array} \right. \Leftrightarrow \min_{f \in \mathcal{DC}\mathcal{H}} J(f)$$

$J, P_i$  and  $R_j$  from  $\mathcal{H} \rightarrow \mathbb{R}$

- questions:

- modelization (transformations)
- optimization theory (existence, uniqueness, characterization of the solution)
- algorithms (no analytical solution)/ implementations

- The importance of being convex:

- convex OP: objective  $J$  and constraints  $P_j$  are convex,  $R_j$  are affine
- why convex? It is the *solvable class*:  
 $\Rightarrow$  the solution exists, is unique and can be found efficiently

# examples of convex OP

## Linear programming LP (standard form)

$$\left\{ \begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \\ \text{and} & \mathbf{x} \succeq 0 \quad (x_i \geq 0, \quad i = 1, n) \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \langle f, k \rangle_{\mathcal{H}} \\ \text{s.t.} & Tf = \mathbf{y} \\ & T: \mathcal{H} \rightarrow \mathbb{R}^n \text{ linear} \end{array} \right.$$

Linear objective and linear constraints

Quadratic programming QP  $\left\{ \begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & \frac{1}{2} \mathbf{x}^\top C \mathbf{x} - \mathbf{d}^\top \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \preceq \mathbf{b} \end{array} \right.$

quadratic objective and linear constraints

## Second order cone programming SOCP

$$\left\{ \begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \|A_i \mathbf{x} - \mathbf{b}_i\|_2 \leq \mathbf{d}_i^\top \mathbf{x} + e_i, \quad i = 1, p \\ \text{and} & F\mathbf{x} = g \end{array} \right. \quad \text{second order cone constraint}$$

# Convex optimization in three slides

- ▶ Initial problem (**primal**): 
$$\begin{cases} \min_{f \in \mathcal{H}} & J(f) & \text{primal objective} \\ \text{s. t.} & P_i(f) \leq 0 & i = 1, p \\ \text{and} & R_j(f) = 0 & j = 1, q \end{cases}$$
- ▶ Lagrangian: 
$$L(f, \alpha, \beta) = J(f) + \sum_{i=1}^p \underbrace{\alpha_i}_{\geq 0} P_i(f) + \sum_{j=1}^q \beta_j R_j(f)$$
- ▶ Lagrange dual function: 
$$Q(\alpha, \beta) = \min_{f \in \mathcal{H}} L(f, \alpha, \beta)$$
  - ▶ Lagrange dual function with linear constraints  $Tf = \mathbf{y}, p = 0$   

$$Q(\alpha) = -\mathbf{y}^\top \alpha - J^*(-T^* \alpha) \quad \text{where } J^* \text{ denotes the conjugate}$$
  - ▶ and for any feasible  $\hat{f}$  (such that  $P_i(\hat{f}) \leq 0$  and  $R_j(\hat{f}) = 0$ )  

$$Q(\alpha, \beta) = \min_{f \in \mathcal{H}} L(f, \alpha, \beta) \leq L(\hat{f}, \alpha, \beta) \leq J(\hat{f})$$
- ▶ Lagrange **dual** problem: 
$$\begin{cases} \max_{\alpha, \beta} & Q(\alpha, \beta) \\ \text{s.t.} & \alpha \preceq 0 \end{cases}$$

# Convex optimization in four slides

Optimality conditions (Karush-Kuhn-Tucker KKT conditions):

$$f^*, \alpha^*, \beta^* \text{ optimum} \Leftrightarrow \begin{cases} \nabla_f J(f^*) + \sum_{i=1}^p \alpha_i^* \nabla_f P_i(f^*) + \sum_{j=1}^q \beta_j^* \nabla_f R_j(f^*) = 0 \\ P_i(f^*) \geq 0 \\ R_j(f^*) = 0 \\ \alpha_i^* \geq 0 \\ \alpha_i^* P_i(f^*) = 0 \end{cases}$$

duality gap:

$$J(f^*) - Q(\alpha^*, \beta^*) = 0;$$

## Optimization summary

1. (try to) transform your problem in a **convex** one (standard form)
2. more variables or more constraints: is the **dual** simpler?
3. solve it: check the KKT conditions

# loss function: the fitting cost

$$\begin{aligned} \ell: \mathcal{H}, \mathcal{X}, \mathcal{Y} &\longrightarrow \mathbb{R}^+ \\ f, \mathbf{x}, y &\longmapsto \ell(y, f(\mathbf{x})) \end{aligned}$$

|             | Regular                                                                                  | singular                                                                                   |
|-------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Convex      | logistic<br>$\log(1 + \exp^{-yf(\mathbf{x})})$<br>L2-gaussian<br>$(f(\mathbf{x}) - y)^2$ | hinge<br>$(yf(\mathbf{x}) - 1)_+$<br>L1-laplacian<br>$ f(\mathbf{x}) - y $                 |
| NON convexe | sigmoid<br>$1 - \tanh(yf(\mathbf{x}))$<br>Cauchy<br>$\log(1 + (f(\mathbf{x}) - y)^2)$    | 0/1<br>$\text{sign}(f(\mathbf{x})) - y$<br>Lp - square root<br>$ f(\mathbf{x}) - y ^{1/2}$ |

$$(x)_+ = \max(x, 0)$$



# regularity criterion: regularization

**Problem:**  $\mathcal{H}$  is too big...

let  $S_0 = \{f \in \mathcal{H} \mid \sum_{i=1}^n \ell(f, \mathbf{x}_i, y_i) = 0\}$

the problem is **ill posed**: solution is not unique.

we have to choose one!  $f_I$  minimal norm solution  $\min_{f \in S_0} \|f\|^2$

regularized: build a **regularization path**

a sequence of problems whose solution  $f_\lambda$  converges towards  $f_I$

3 way to do it:

1. penalization  $R(f): \min_{f \in \mathcal{H}} \left( \sum_{i=1}^n \ell(f, \mathbf{x}_i, y_i), R(f) \right)$

2. subspace  $\mathcal{H}_1 \subset \dots \mathcal{H}_k \dots \subset \mathcal{H} : \min_{f \in \mathcal{H}_k} \sum_{i=1}^n \ell(f, \mathbf{x}_i, y_i)$

3. iterative approach: gradient (Landweber-Friedman) or conjugate gradient (Krylov subspace)

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# penalization choice

|            | regular                                                       | singular                            |
|------------|---------------------------------------------------------------|-------------------------------------|
| convex     | $\mathcal{L}_2$<br>$\ f\ _2^2$                                | $\mathcal{L}_1$<br>$\ f\ _1$        |
| non convex | normalized-SCAD<br>$\sum_j \frac{\alpha_j^2}{1 + \alpha_j^2}$ | $\mathcal{L}_p, p < 1$<br>$\ f\ _p$ |

$$\|f\|_{\mathcal{H}}^2 = \alpha^\top K \alpha \quad f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i) \quad \|f\|_1 = \sum_{i=1}^n |\alpha_i|,$$

$$\|f\|_p^p = \sum_{i=1}^n |\alpha_i|^p$$

no more convex when  $p < 1$

## an attempt at classifying some Kernel learning algorithm

| $\ell$ | $h(\mathcal{H})$      | Regression        | Classification                    |
|--------|-----------------------|-------------------|-----------------------------------|
| sing.  | sing. $\mathcal{L}_1$ | K Danzig Selector | LP SVM                            |
| reg.   | sing. $\mathcal{L}_1$ | K LASSO<br>K LARS | K reg. log. L1                    |
| sing.  | reg. $\mathcal{L}_2$  | SVR               | SVM                               |
| reg.   | reg. $\mathcal{L}_2$  | Splines           | K-logistic reg.<br>Lagrangian SVM |

**Table:** SVM and SVR stand for *support vector machine* and *support vector regression*. LP *linear programming*, LARS *Least angle regression stagewise* and reg. log. *logistic regrssion*. K. represent the kernelize version o fthe linear algorithms..







# let's summarize

- ▶ hypothesis  $k$  et  $\mathcal{H}$
- ▶ data fidelity (loss):  $\ell$  convex
- ▶ regularity: learning as a multicriterion optimization
- ▶  $\ell_1$  = sparsity + convexity = some efficiency



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